Unit 4 Social Issues and the Environment

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- **From Unsustainable to Sustainable Development**
  - Consumerism and waste products.
- **Urban problems**
  - Related to energy,
  - Related to water: water conservation, rain water harvesting, watershed management
  - Resettlement and rehabilitation of people
  - Wasteland reclamation.
- **Climate change and Global issues**: Global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust.
- **Environmental Law**
- **Environmental ethics**: Issues and possible solutions

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Sustainability and Circular Economy

Timeline towards attaining circular economy

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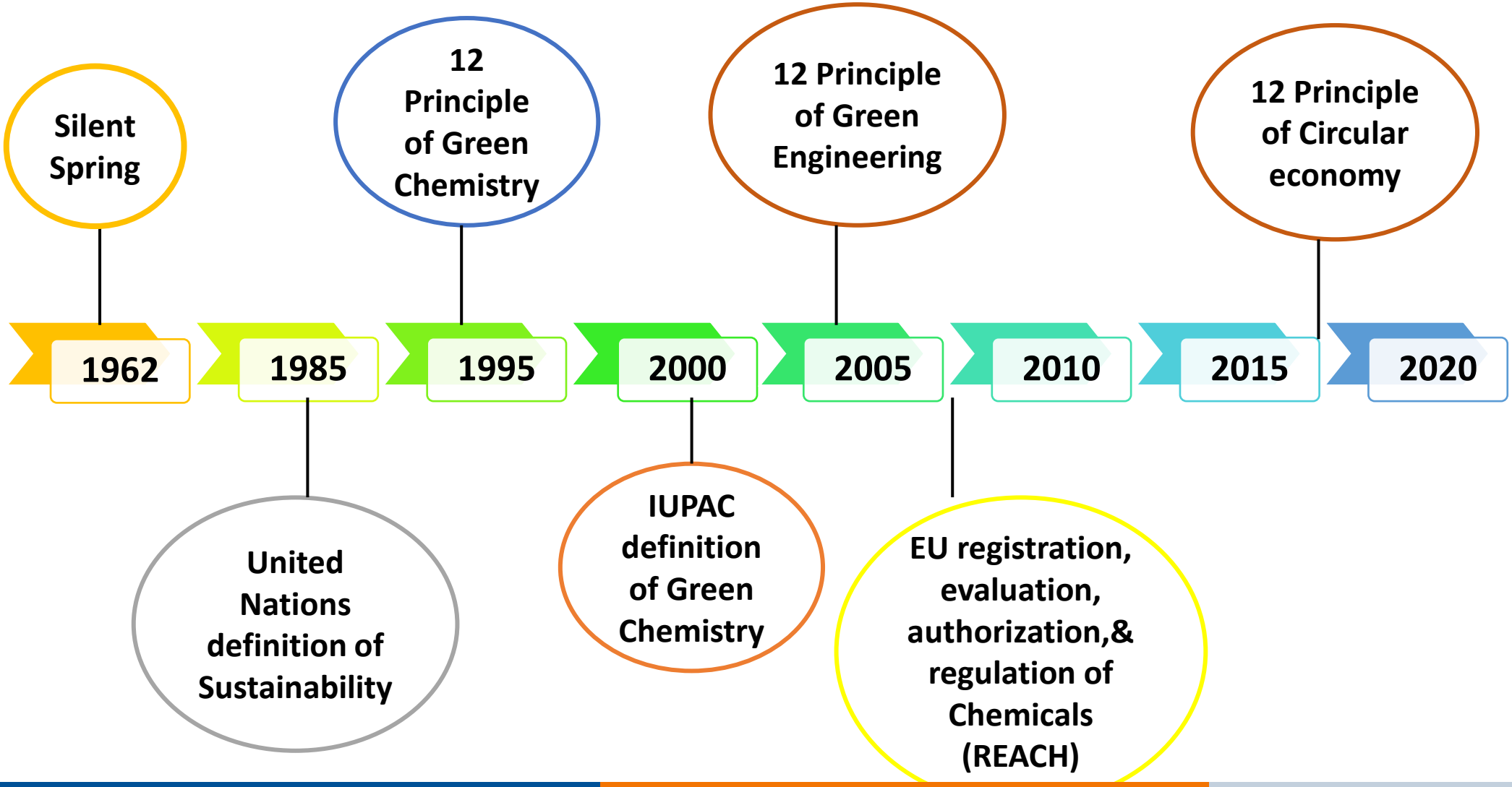

Linear Economy Approach

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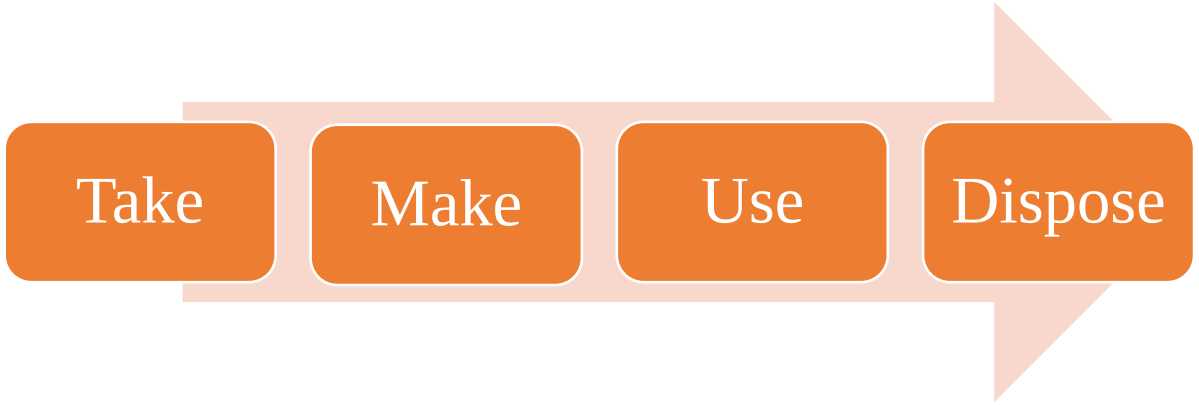
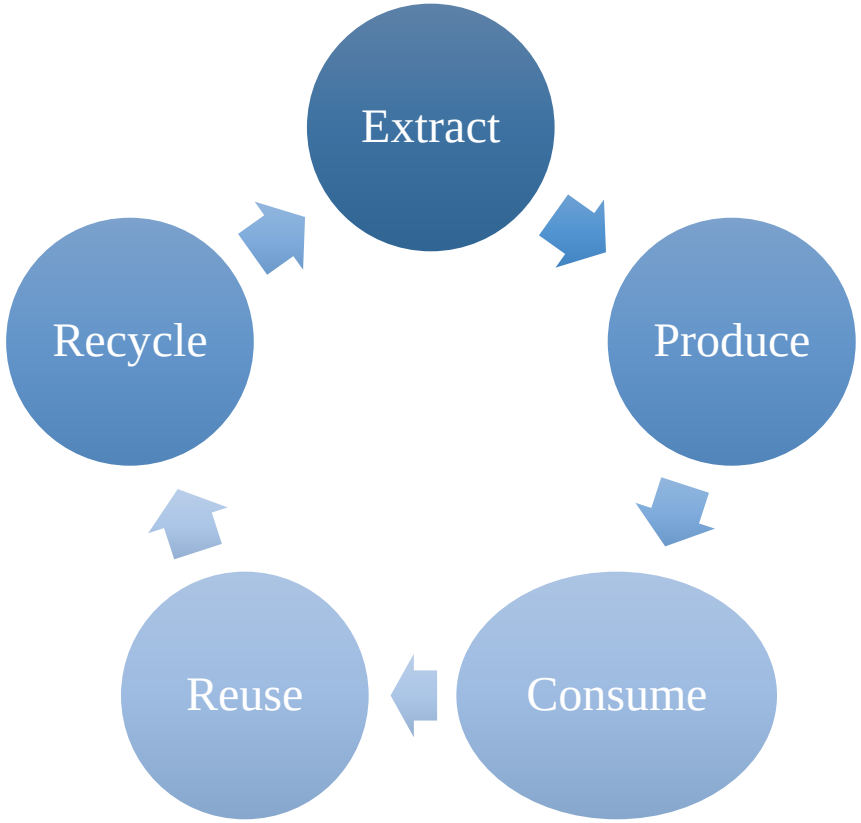
- The materials economy of today is largely linear - “take-make-use-dispose.”
- Products and materials not used to their full potential - move in one direction – from raw material to waste.
- The industrial revolution laid the framework - goods were able to be mass produced.
- This global development was only possible by extracting finite resources and degrading natural capital to generate economic value.
- A transition to a circular economy will be crucial to allow us to thrive in the future.



What is a circular economy?

- European Action Plan states that circular economy is an economy *'where the value of products, materials and resources is maintained in the economy for as long as possible, and the generation of waste minimized'*
- **An economy that is restorative and regenerative by design.**
- Tackles climate change and other global challenges like biodiversity loss, waste, and pollution, by decoupling economic activity from the consumption of finite resources.



Key principles of the circular economy

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1. **Design for longevity:** Products are designed to be durable, repairable, and upgradeable.
2. **Resource efficiency:** Minimizing the use of raw materials and energy during production and consumption processes to reduce waste and pollution.
3. **Reuse:** Encouraging the reuse of products or components for their original purpose or a different application.
4. **Repair and maintenance:** Promoting repair services and easy access to spare parts to extend the lifespan of products.
5. **Remanufacturing:** Refurbishing or restoring used products to "like-new" condition, reducing the need for new manufacturing.
6. **Recycling:** Recovering materials from discarded products and incorporating them back into the production process to create new products.
7. **Biomimicry:** Taking inspiration from natural systems and processes to design products and technologies that are inherently regenerative and sustainable

Does Circularity ensures Sustainability?

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- Brundtland Report (1987): *"Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs"*

- Sustainability implies a wider approach than simply economic growth, as it includes other indicators as social and environmental ones.

- CE activities should be assessed in the light of economic, environmental, and social sustainability.
- The sustainability of CE strategies needs to be **measured against their linear counterparts** to identify and **avoid** strategies that increase circularity but lead to **unintended externalities.**

Need of Sustainability check as Circularity always may not be the right choice due to:

- *Dissipative use of much of the currently used resources,*
- *Need of virgin materials can't always be offset by recycling,*
- *Commingled wastes are always difficult to recycle*

NOTE : Recycling some materials creates greater environmental impact than virgin materials because of high energy requirements, toxicity concerns

United Nations Sustainable Development Goals

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Unsustainable → Sustainable

Defined as meeting the needs of the present generation without compromising the ability of future generations to meet their own needs.

Reduce Environmental Footprint
Green products
non-toxic, long-life, recyclable
Cleaner production
using fewer resources
Generate Increased Income
Circular Economy
Recycle waste, reuse resources
Better service
to extend lifespan
Minimize Waste
Collect at end-of-life, remanufacture
Reduce Resource Dependency

URBAN PROBLEM RELATED TO ENERGY

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- Now 50%population lives in Urban areas
- **Urban sprawl**: Uncontrollable and unplanned growth
- Densely populated, consume more resources, NEED MORE ENERGY

- **Energy demanding activities**
  - Residential and Commercial lighting- Malls, offices, hotels.
  - Private and Public transport: EV vehicles
  - Modern lifestyle: electronic gadgets.
  - Industries
  - Waste disposal, Prevention and Control of pollution

Consequence

- Unequal distribution of energy
- Power cuts and load – shedding
- Economics

WATER CONSERVATION

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Rainwater harvesting: is a technique of collecting and storing rainwater and increasing the recharge of ground water by collected water.

Why harvest rainwater?

- to meet the growing water requirements
- to raise the water table by recharging ground water
- to reduces soil erosion as surface run-off is reduced
- to avoid flooding of roads
- to reduce groundwater contamination
- to mitigates the effect of drought
- to decrease salinity (by dilution)

5 Principles of Rain water Harvesting

Use of Harvested Rainwater

1. Provides a source of water at the point where it is needed.
2. Provides an essential reserve in times of emergency
3. Construction of a rooftop rainwater catchment system is simple, cheap.
4. Recharge aquifers
5. Improve soil moisture content
6. Rainwater may be superior to those of groundwater sometimes
7. Construction, operation, and maintenance are not labor-intensive.

Disadvantages of Rainwater harvesting

1. Depends upon the frequency and amount of rainfall
2. Low storage capacities will limit rainwater harvesting
3. Increased storage capacities add to construction and operating costs.
4. Rainfall harvesting systems increase construction costs.

WATERSHED MANAGEMENT

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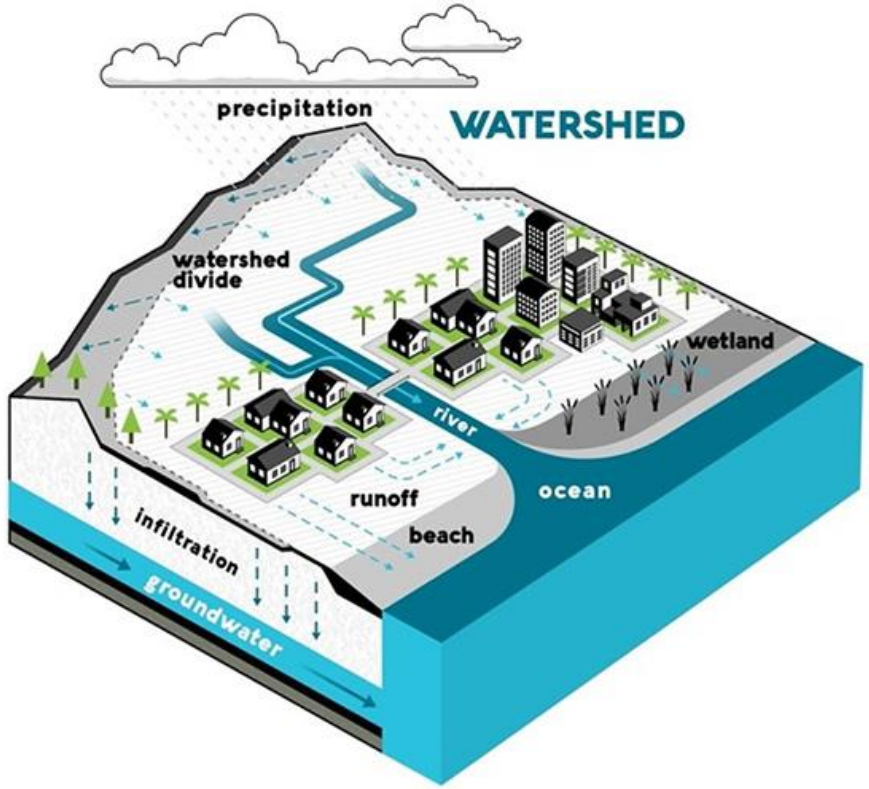
• A watershed is a geographic area through which water flows across the land and drains into a common body of water, whether a stream, river, lake, or ocean.



Objectives of watershed management

- To use watershed for domestic water supply, irrigation, hydropower generation etc.
- To use watershed for minimizing soil erosion and moisture retention.
- To minimize the risks of floods, droughts and landslides. Importance of watershed management
- Recharges the groundwater table.
- Restores soil fertility and helps in soil conservation
- Promotes sustainable agriculture.
- Protects biodiversity of a region,

Challenge in Resettlement and Rehabilitation in India

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Often arising in the context of infrastructure development projects such as dams, highways, urban development, and industrial expansion.

1. **Involuntary Displacement**
2. **Loss of Livelihood**
3. **Inadequate Compensation**
4. **Lack of Land Titles**
5. **Access to Basic Services**
6. **Cultural Disruption**
7. **Social Fragmentation**
8. **Delay in Rehabilitation**:

Proper resettlement and rehabilitation

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1. Baseline Survey and Impact Assessment
2. Compensation and Entitlements
3. Resettlement Planning
4. Livelihood Restoration
5. Housing and Infrastructure Development
6. Social and Cultural Support
7. Access to Basic Services
8. Legal Support
9. Monitoring and Evaluation
10. Grievance Redressal
11. Sustainability and Self-Reliance
12. Long-Term Support
13. Documentation and Reporting
14. Stakeholder Collaboration

Waste land Reclamation

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- Defined as **economically unproductive land** suffering from environmental deterioration. The lands are unstable in ecologically and topographically with complete loss of its fertility status
- Unhabitated and uncultivated land which **no longer serves any purposes**
- Most wastelands include:
  - **Salt Affected Land**
  - **Salty Area**
  - **Gullied Area**
  - **Undulating Uplands**
  - **Barren Hill Ridge**
  - **Snow Covered and Glacial Areas**
  - **Abandoned Cultivation Land**
- Rajasthan- maximum wasteland is 36million ha
- Harayana wastelands cover 8.4% of the total area which comprises saline, sodic or sandy land areas

Formation of wasteland

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Natural processes

- Torrential rain rapid flow of water along slopes
- Run-off
- Wave action and melting
- movement of snow
  Stormy wind, cyclone

Anthropogenic activities

- Deforestation
- Overgrazing
- Erroneous agricultural practices
- Mining

Types of wasteland

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1. Water logged land

Presence of water table at or near the land surface creates water logged lands. This water stands through out the year and the group get saturated with water

2. Ravine or Gullies

Formed by erosion of running water .These gullies are generally run parallel to each other and end in rivers. **Riverine lands**

3. Riverine

is the initial stage of Ravines or gullies, formed due to surface runoff

4. Undulation land

5. Coastal sand dunes

Formed by accumulation of sand in the coastal areas either from the sand available in the same coastal area or transported by water erosion in other areas.

6. Sand and wind eroded lands

Formed by accumulation of sand through wind erosion in hot desert areas

7. Degraded forest land

Formed by total destruction of vegetation in a forest area

8. Mining waste lands

Formed by dumping and disposal of mines wastes

9. Industrial waste land

Disposal of liquid and solid wastes of industries

10. Banjar lands

Suitable for cultivation but presently out of cultivation for a period of more than 5yrs

11. Old Fallow lands

Suitable for cultivation but presently out of cultivation for period of not less than 2 yrs

12. Fallow lands

Taken up cultivation but out of cultivation for more than one year

Wasteland development board

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Objectives

- To improve the physical structure and quality of the soil
- To improve the availability of water
- To prevent soil erosion, flooding and landslides
- To conserve the biological resources of the land for sustainable use

Reclamation processes

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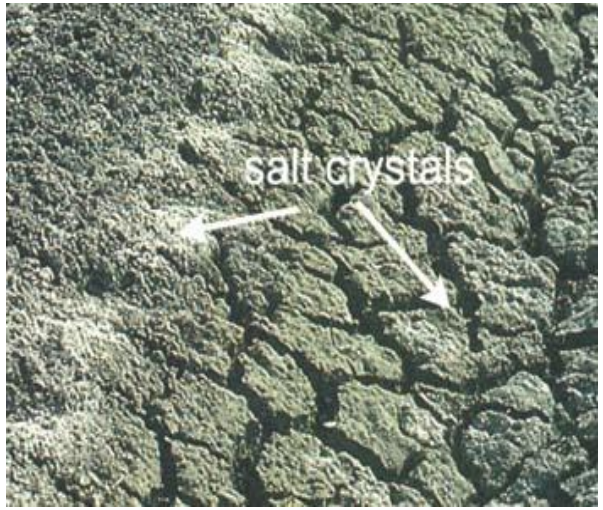
1. **Land Development and Leaching**
   - Salinity of the land is removed by leaching through application of water.
   - Intermittent sprinkling reduces 90% salinity in upper 60cm layer.



2. **Drainage**
   - Excess water is removed by artificial drainage methods.
   - Both surface and sub-surface drainage methods are used.

3. **Irrigation**
   - Surface irrigation with precise land leveling ,smoothening and efficient hydraulic design help reduce water logging and salinity.
   - Thin and frequent irrigation with controlled amount of water

4. Crop Rotation and Selection

- Tolerance of crops to salts plays a major role to start cultivation of a particular crop in certain area.
- Water necessity and the salinity tolerance are considered and cultivation is started.

5. Green manures and fertilizers

- Farm yard manures or nitrogen fertilizers reduce salinity
- Blue Green algae are shown to reduce salinity to promising levels.
- Gypsum amendment : Ca of gypsum replaces Na from sodic salts by ion-exchange method
- Sulphuric acid and Fe & Al sulphate neutralizes the alkalinity of the soil

6. Afforestation and Social Forestry: strip plantation on road, rail and canal sides, degraded wasteland etc

7. Environmental biotechnology approach : Micropropagation, use of mycorrhizae, use nitrogen fixing microbes

CLIMATE CHANGE

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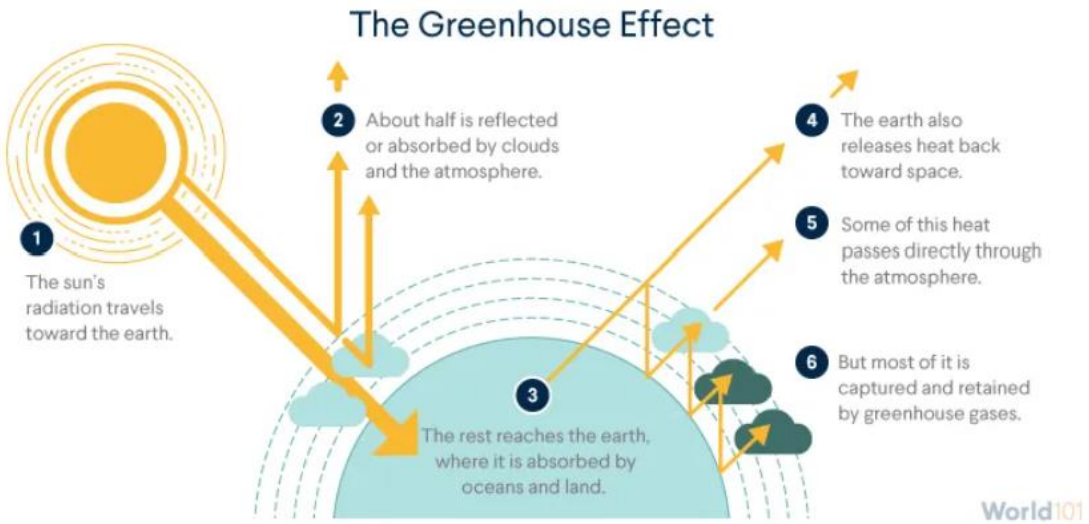
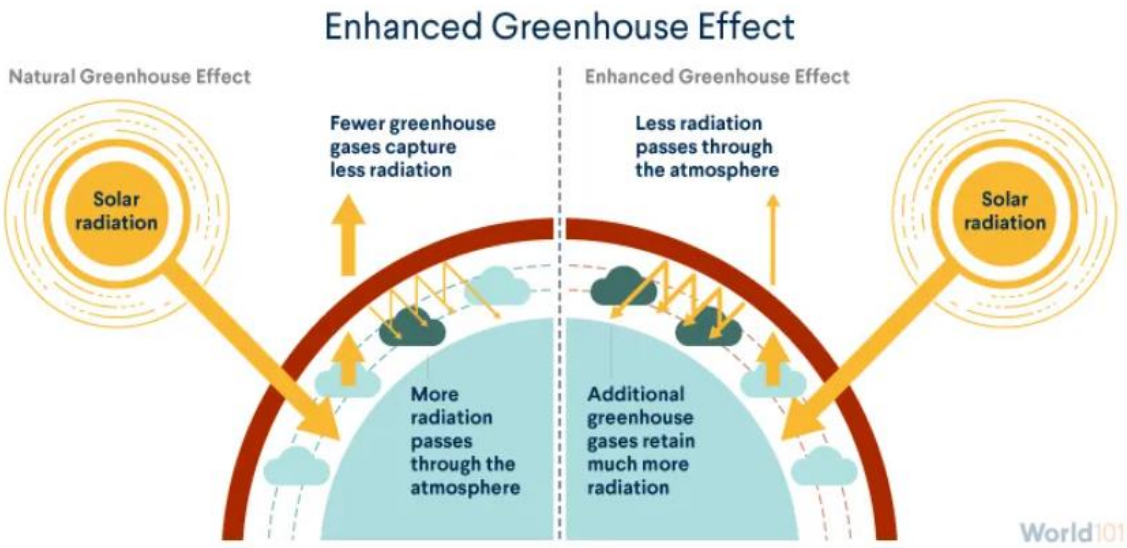
- **Climate** = long-term atmospheric conditions
- **Weather** = short-term atmospheric conditions
- Both climate and weather are dynamic – they change with time
- ***Increase in temperature***
- **Greenhouse**- is a structure with walls and roof made chiefly of transparent material like glass, in which plants requiring regulated climatic conditions are grown.

- **Greenhouse effect is the process by which radiations from the sun are absorbed by the greenhouse gases and not reflected back into space. This insulates the surface of the earth and prevents it from freezing.**

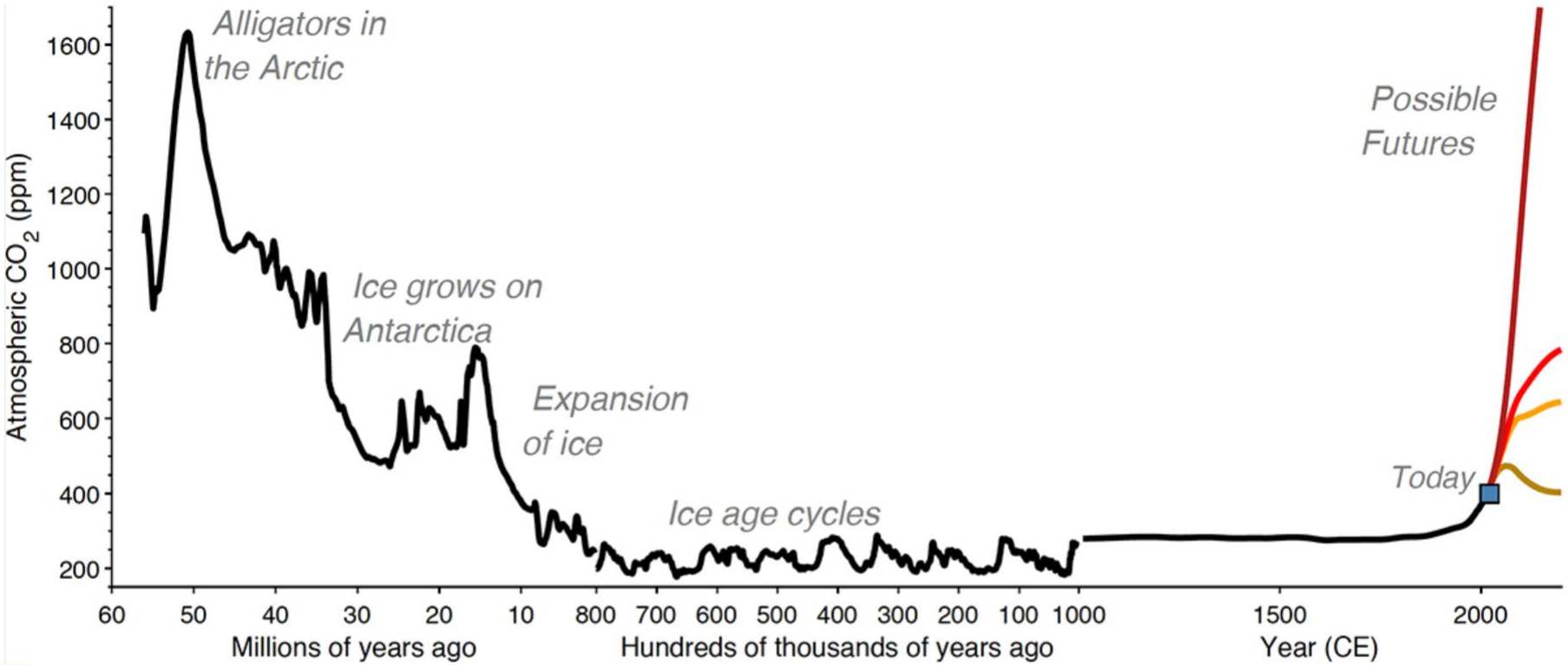


GHGs	Residence time (years)	GWP	Sources
Carbon dioxide	Variable	1	
Water vapor	Variable	1	
Methane	12.2+/-3	21	
Nitrous dioxide	120	310	Fertilizers, industries and combustion
CFCs- 12	102	6200-7100	Liquid coolant and foam
HCFC-22	12.1	1300-1400	Liquid coolant
CF4	50, 000	6500	Aluminium production
SF4	3200	23900	Dielectric fluid



Consequences of greenhouse effect

- Increase in temperature
- Sea level rise: coastal hazards and biodiversity Global Weather Patterns: Typhoons and hurricanes, El nino
- Human Related Problems: Heat stroke cases, Mosquitoes, vector for disease
- Crop yield



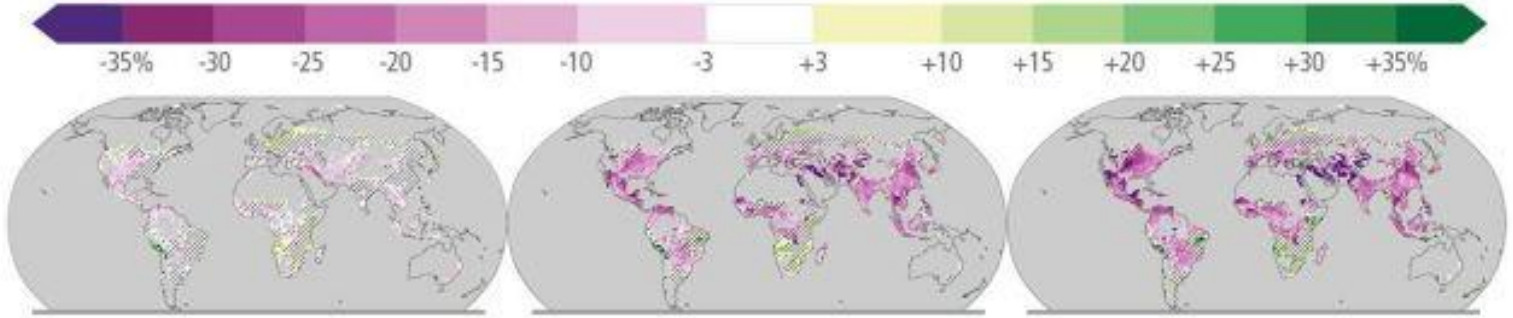
At CO2 levels of around 1500 ppm, last seen about 50 million years ago, it was so warm that we find fossilized alligators in the Arctic, researchers say.

Courtesy of James Rae

Future climate change is projected to increase the severity of impacts across natural and human systems and will increase regional differences

Examples of impacts without additional adaptation

Food production impacts

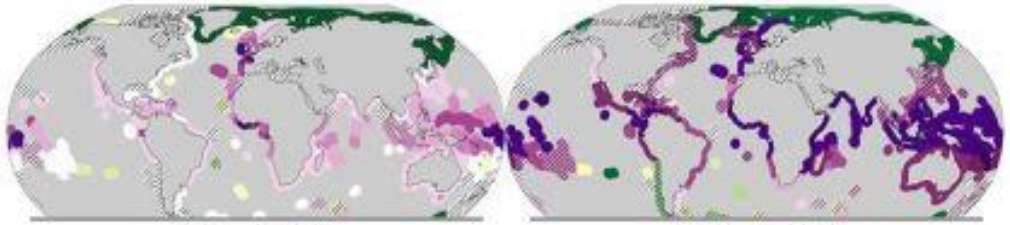


1) **Maize yield**[4]
Changes (%) in yield

1.6 – 2.4°C **3.3 – 4.8°C** **3.9 – 6.0°C**

[4]Projected regional impacts reflect biophysical responses to changing temperature, precipitation, solar radiation, humidity, wind, and CO_2 enhancement of growth and water retention in currently cultivated areas. Models assume that irrigated areas are not water-limited. Models do not represent pests, diseases, future agro-technological changes and some extreme climate responses.

2) **Fisheries yield**[5]
Changes (%) in maximum catch potential



0.9 – 2.0°C **3.4 – 5.2°C**

Areas with little or no production, or not assessed

Areas with model disagreement

[5]Projected regional impacts reflect fisheries and marine ecosystem responses to ocean physical and biogeochemical conditions such as temperature, oxygen level and net primary production. Models do not represent changes in fishing activities and some extreme climatic conditions. Projected changes in thea Arctic regions have low confidence due to uncertainties associated with modelling multiple interacting drivers and ecosystem responses.

Source: IPCC, 2023

KYOTO PROTOCOL

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- Climate Change Convention agreed at the **Rio Earth Summit in 1992**
- To stabilise their emission of **CO2 at 1990 level by the year 2000**.
- Argued that flexibility would allow countries to emit more greenhouse gases if they planted more trees.
- Could demonstrate that they had reduced pollution in other countries.
- Kyoto protocol, **rich industrial countries agreed to reduce their emissions of greenhouse gases by an average of 5.2 per cent by 2010**.
- The main objective of the Kyoto Protocol – stablising global climate at non dangerous level.

Intergovernmental panel on climate change(IPCC)

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- World Meteorological Organisation(WMO) and the United Nation Environmental Program (UNEP), 1988
  - To access available information on science of climate change
  - Scientific understanding of past and present climate of climate variability.
  - Predictability and climate change modelling
  - Mitigation of climate change
- **On December 12, 2015, in Paris, 195 to lower greenhouse gas emissions aims to prevent global temperatures post revolution temperatures by 2˚C. from rising above pre–Industrial**
  - ***Low-growth scenario: concentration of CO2 of 450ppm by 2100***
  - ***High-growth scenario: concentration of CO2 of 900ppm by 2100***
  - ***Middle-growth scenario: concentration of CO2 of 700ppm by 2100***

GLOBAL OZONE PROBLEM

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- Formation of Ozone
- Main ozone factory is at the tropical altitude (30-40 km altitude), but it is transported to poles (North pole Summer and South pole in October)
- **Consequence of ozone depletion:**UV-C is lethal to microorganism

- **OZONE DEPLETION PROCESS**
  - **Hydrogen system (OH): 10%, above 40 km of surface**
  - **Nitrogen system (N2O): 60 %**
  - **Chlorine system (CFCs)**

SEQUENCE OF PROCESSES FORMING THE OZONE HOLE

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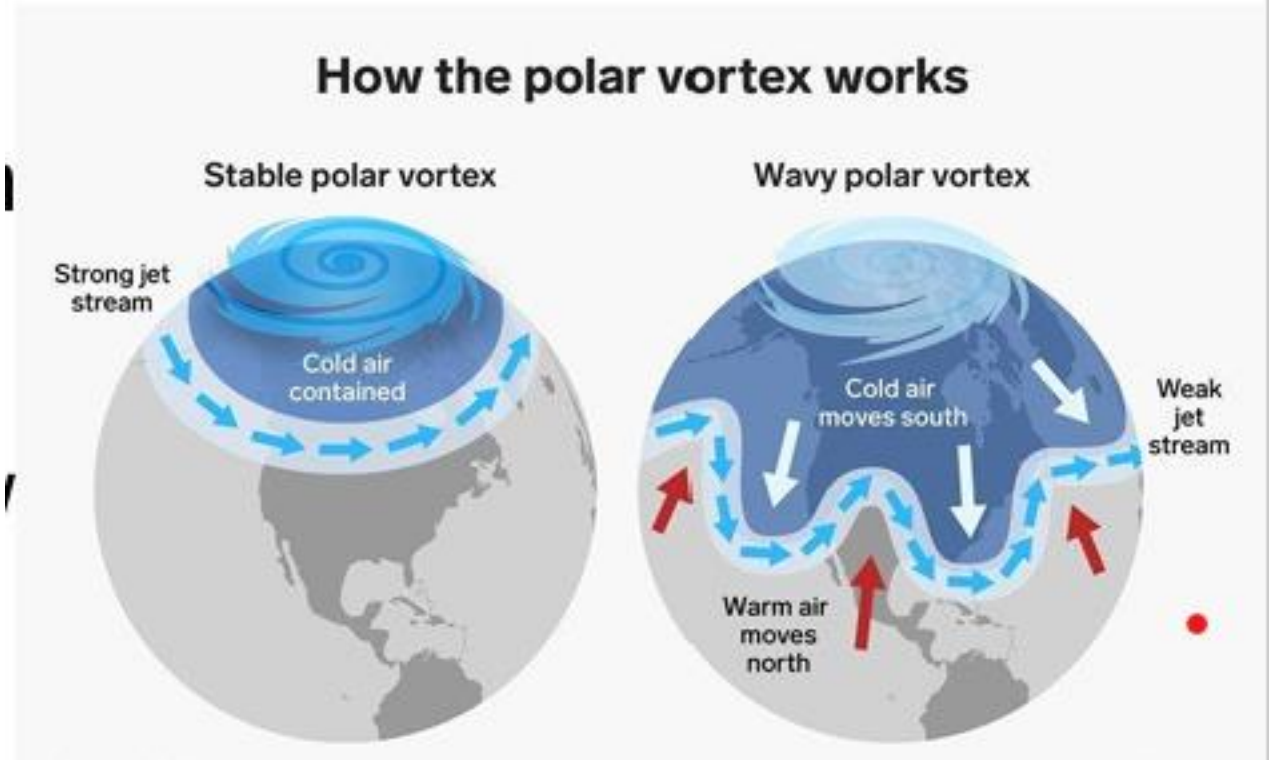
OZONE HOLE: ANTRACTICA: Polar stratospheric clouds

- Vortex formation
- Very low temperature
- De-humidification de-nitrification
- Pre-conditioning
- Spring arrives sunlight
- Ozone destruction by catalysis



THE MONTREAL PROTOCOL: banning of CFCs

ACID RAIN

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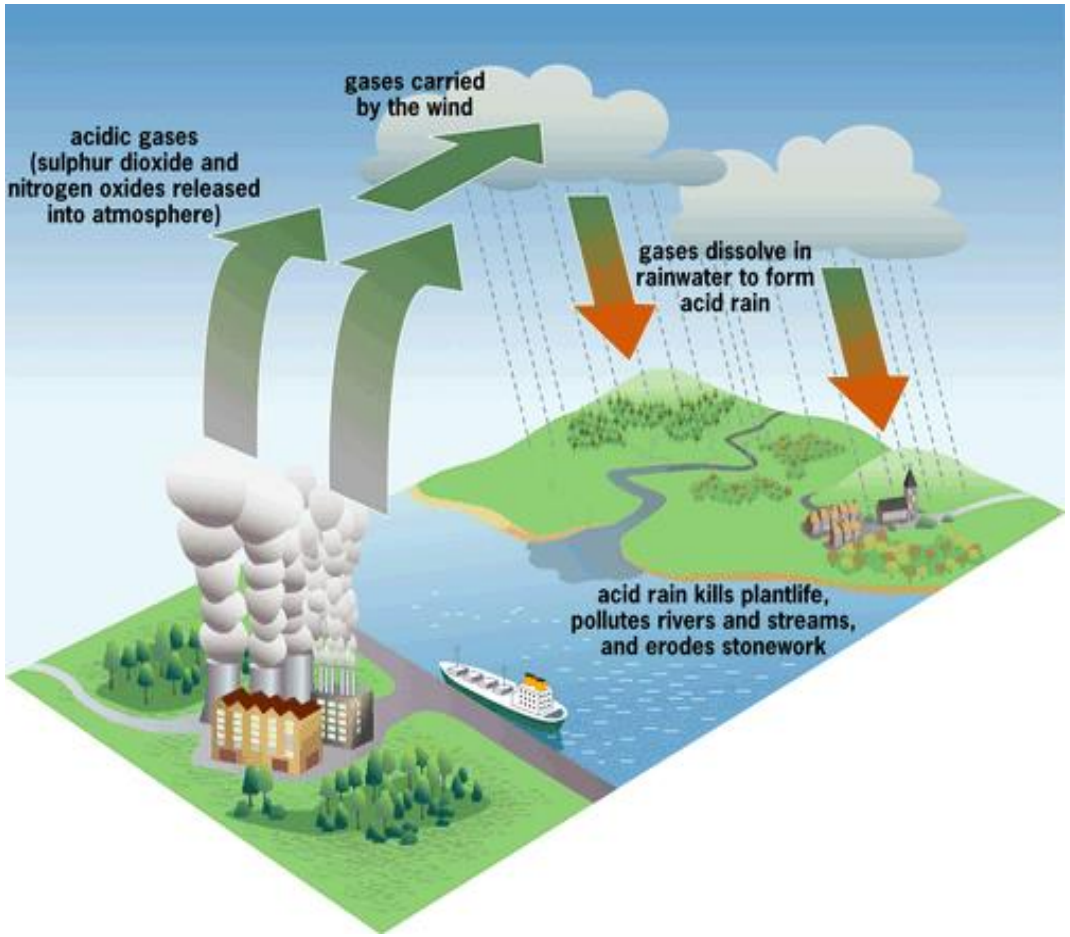
- **Oxides of sulphur and nitrogen** from fossil fuel combustion and industrial operations are the major sources of acid making gases.

- These gases combine with water moisture in the atmosphere as a result of which sulphuric acid (H_2SO_4) and nitric acid (HNO_3) are formed.

- **These chemical can be deposited on particulate matter (dry deposition) or on raindrop, snow, dew (wet- deposition)**

- Acid rains corrode buildings, badly affects aquatic life and also cause premature deaths and cancer in humans and animals.



CLIMATE CHANGE HAZARDS/RISKS

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PRIMARY THREAT	SECONDARY THREAT
Extreme precipitation events within short spell	Flooding in river basin Increased erosion and landslide
Cyclones/Hurricanes/Strom	Strom surges High winds/Strom damage
Rising sea level	Costal and fluvial flooding Saline penetration in fresh water aquifers
Increased temperature	Heat waves Rise of drought incidence Loss of glaciers Loss of permafrost sea ice Incidence of natural bush fire
Ecosystem adaptation	Human adaptation/migration Distribution of disease incidence high

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ENVIRONMENTAL ETHICS AND LAW

Environmental Ethics

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It seeks to address the ethical challenges posed by environmental issues, propose solutions, and assess why environmental ethics has faced both successes and failures.

Challenges in Environmental Ethics:

1. Anthropocentrism vs. Biocentrism
2. Environmental Degradation
3. Climate Change

Solutions to Environmental Ethics Challenges:

1. Environmental Education
2. Legislation and Regulation
3. Sustainable Practices
4. Ethical Stewardship

Failures in Environmental Ethics:

1. Short-Term Interests
2. Lack of Enforcement
3. Value Conflicts
4. Global Cooperation

Environmental Economics

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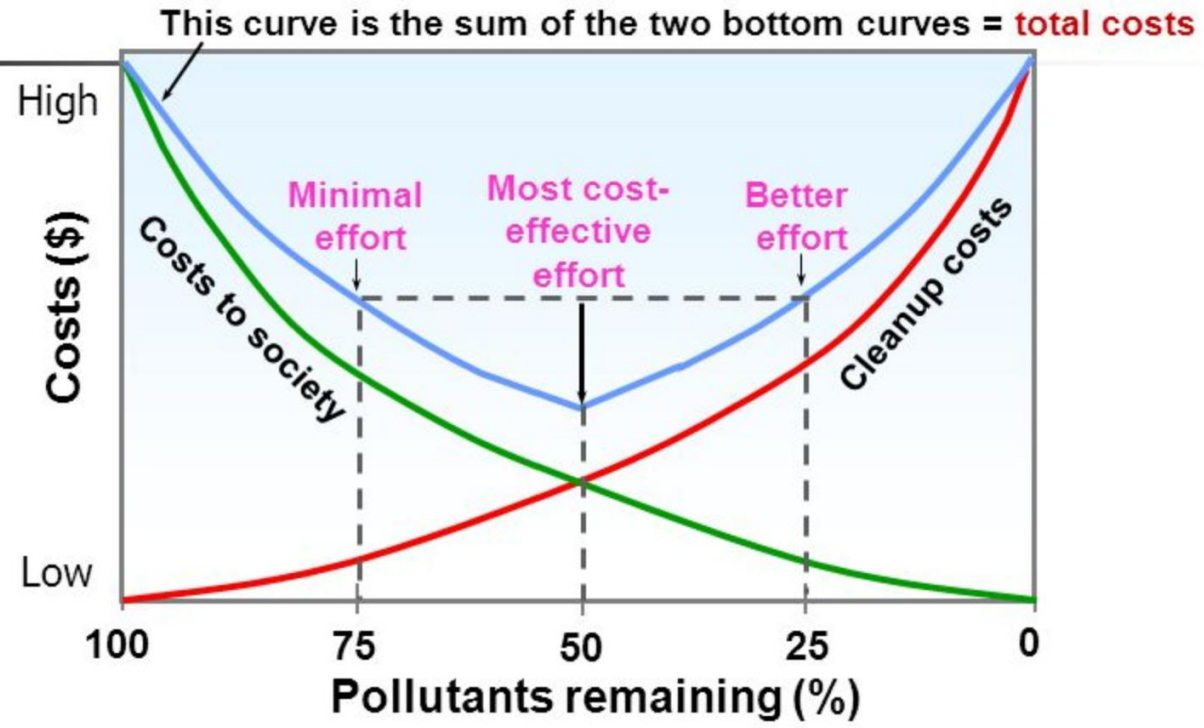
- Earth Capital or Natural Resources
- Manufactured Capital
- Human Capital

Cost – effective analysis for optimal level of pollution

Optimum Level Of Pollution



To find the break-even point, economists plot (1) estimated costs of cleaning up and (2) estimated social costs of pollution (harmful external costs of pollution to society), then add the two curves to get a third curve - the total costs. Lowest point on the third curve (the breakeven point) = optimum level of pollution. This graph shows the optimum level at 50%, but the actual level varies with the pollutant. *(See Fig. 2-7, Miller (2003))*

Water (Prevention and Control of Pollution) Act, 1974

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1. Background:

- The Act was enacted in **response to the increasing water pollution challenges in India.**
- **It was amended in 1988 to align with the Water Act with the provisions of the Environment (Protection) Act, 1986.**

2. Objectives:

- To prevent and control water pollution.
- To maintain or restore the wholesomeness of water.

3. Regulatory Authorities:

- The Act established **State Pollution Control Boards and the Central Pollution Control Board** to enforce its provisions.
- These bodies are responsible for issuing **permits, setting effluent standards, and conducting water quality monitoring.**

4. Key Provisions:

a) **Consent to Establish and Operate:** - Any person or entity that wishes to establish or operate an industry, process, or sewerage system is required to obtain consent from the relevant Pollution Control Board.

b) **Effluent Standards:** - The Act empowers regulatory authorities to specify standards for the discharge of pollutants into water bodies. - These standards vary depending on the class of water, the type of industry, and the intended use of the water.

c) **Monitoring and Inspection:** - Regulatory bodies conduct regular monitoring and inspections to ensure compliance with the Act. - The Act allows for the collection of samples and the assessment of water quality.

d) **Penalties and Offenses:** - The Act prescribes penalties for non-compliance, including fines and imprisonment. - It categorizes offenses based on the severity of pollution.

e) **Appeals:** - Provisions for appealing against decisions and orders of Pollution Control Boards are included in the Act. - These appeals can be made to the Appellate Authority, if available.

f) **Emergency Measures:** - In cases of emergencies, regulatory authorities can take immediate measures to prevent pollution. - The responsible entity will be liable for the costs incurred.

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5. Industries Covered:

- The Act applies to a wide range of industries and processes, including manufacturing, mining, thermal power plants, and more.
- **Industries are classified into red, orange, and green categories based on their pollution potential.**

6. Amendment in 1988:

- It expanded the scope of the **Act to include hazardous substances.**

7. Impact and Challenges:

- The Act has played a crucial role in curbing water pollution in India.
- Challenges include **enforcement issues, lack of monitoring infrastructure, and emerging contaminants**.

Wildlife Protection Act, 1972

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1. Objectives:

- Protect wildlife from hunting and poaching.
- Regulate trade in wildlife and their derivatives.
- Promote wildlife conservation and habitat preservation.

2. Key Provisions:

a. **Protected Species and Habitats:** - Various schedules, providing different levels of protection. - It designates wildlife sanctuaries, national parks, and protected areas to safeguard habitats.

b. **Prohibition on Hunting:** - The Act prohibits the hunting of any wild animal specified in Schedules I to IV. - Exceptions are made for certain situations, such as scientific research and defense of human life.

c. **Regulation of Trade:** - The Act regulates the trade of wildlife and their derivatives through permits and licenses. - Trade in species listed under Schedule I is generally prohibited.

d. **National Parks and Sanctuaries:** - National parks are areas set aside for the conservation of wildlife where all human activities are regulated. - Sanctuaries have some degree of human activity, but these are also regulated to protect wildlife.

e. **Capture and Transport:** - Permits are required for capturing or transporting wildlife. - These permits are issued only for specified purposes, such as scientific research or public exhibition.

f. **Penalties and Enforcement:** - The Act prescribes penalties for violations, including fines and imprisonment. - The enforcement of the Act is overseen by wildlife authorities.

3. Role of the Central Zoo Authority (CZA):

•The CZA was established to oversee the operation of zoos and to ensure that they meet prescribed standards for the upkeep and welfare of captive animals.

4. Impact:

•The Wildlife Protection Act, 1972 has played a significant role in conserving wildlife and their habitats in India.

•It has contributed to the protection of endangered species and the establishment of a network of protected areas.

•Challenges include addressing human-wildlife conflicts and strengthening enforcement

Forest (Conservation) Act, 1980

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Objectives:

- **Conservation of Forests**
- **Regulation of Forest Land**
- **Preservation of Biodiversity**

Key Provisions:

a) **Restrictions on Diversion of Forest Land:** forest land cannot be diverted for non-forest purposes, such as mining, industry, or infrastructure development, without prior approval from the central government

b) **Prior Approval:** Any proposal to divert forest land must be submitted to the **central government**, which reviews the proposal and may grant or deny approval based on environmental considerations.

c) **National Forest Policy:** The Act aligns with the **National Forest Policy of 1988, which emphasizes the ecological and environmental significance of forests and sustainable forest management**.

d) **Penalties:** The Act specifies penalties for contravention, which can include fines and imprisonment.

e) **Biodiversity Conservation:** The Act indirectly contributes to biodiversity conservation by protecting forests, which are crucial for diverse plant and animal species.

f) **Exceptions: The Act provides exceptions for projects related to defense, national security, and certain other essential purposes, although these are also subject to environmental safeguards**.

Impact:

- It aligns with India's commitment to preserving its rich biodiversity and maintaining ecological balance.
- Challenges include balancing economic development with environmental conservation.

Environment (Protection) Act, 1986

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1. Objectives:

- Protect and improve the quality of the environment.
- Prevent, control, and abate environmental pollution.
- Set standards for emissions and discharges of pollutants into the environment.

2. Key Provisions:

a) **Regulatory Authorities:** - The Act empowers the central government to take measures to protect the environment.

b) **Environmental Standards:** - The Act provides the authority to **the central government to set environmental standards for various industries and processes. - These standards cover emissions, discharges, noise levels, and other environmental aspects.**

c) **Regulation of Hazardous Substances:** - The Act includes provisions for the regulation of hazardous substances and handling procedures. - It covers the import, export, transportation, and storage of hazardous materials.

d) **Environmental Impact Assessment (EIA):** - The Act includes provisions for environmental impact assessments for new projects or activities. - Prior environmental clearance is required for activities that can have a significant impact on the environment.

e) **Penalties and Enforcement:** - The Act prescribes penalties for violations, including fines and imprisonment. - Regulatory authorities, including the State Pollution Control Boards, are responsible for enforcement.

3. Hazardous Substances and Activities:

- The Act includes provisions for regulating activities and substances that have the potential to cause environmental harm.
- These regulations encompass industries like mining, manufacturing, and hazardous waste management.

4. Impact:

- The Environment (Protection) Act, 1986 has played a crucial role in establishing environmental standards and regulations in India.
- It has contributed to improved air and water quality, the prevention of pollution, and better management of hazardous substances.
- Challenges include effective enforcement and addressing emerging environmental issues

Biological Diversity Act, 2002

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Objectives:

- Conserve biological diversity and ensure sustainable use of its components.
- Regulate access to biological resources and associated traditional knowledge.
- Promote fair and equitable sharing of benefits arising from the use of these resources and knowledge.

Key Provisions:

1. **Regulation of Biological Resources:** regulates access to biological resources by requiring prior approval for their utilization for commercial purposes.
2. **Biodiversity Management Committees (BMCs):** to assist in biodiversity conservation and sustainable use.
3. **Access and Benefit-Sharing (ABS):** procedures for seeking access to biological resources and associated traditional knowledge. It also establishes mechanisms for benefit-sharing with local communities and stakeholders.
4. **Protected Areas:** declaration of areas as biodiversity heritage sites
5. **Intellectual Property Rights (IPR):** patents derived from biological resources and traditional knowledge
6. **Compliance:**
    1. Entities seeking access to biological resources must comply with ABS provisions.
    2. The Act imposes penalties for non-compliance, including fines and imprisonment.
7. **National Biodiversity Authority (NBA):**
8. **State Biodiversity Boards (SBBs): Biodiversity Fund:** to support biodiversity conservation activities.

National Green Tribunal (NGT) Act, 2010

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Objectives:

- Provide effective and expeditious disposal of cases related to environmental protection and conservation.
- Address environmental disputes and violations by enforcing environmental laws.

Key Provisions:

1. **Establishment of the National Green Tribunal:** as a specialized judicial body to handle environmental cases. The Tribunal consists of judicial and expert members with expertise in various aspects of environmental law.
2. **Jurisdiction:** The NGT has jurisdiction over cases related to the environmental matters.
3. **Filing of Cases:** The Act allows individuals and organizations to file cases directly with the NGT, without the need for extensive legal procedures.
4. **Advisory and Expertise:** The NGT can seek advice and expertise from scientists, experts.
5. **Penalties:** The NGT has the authority to impose penalties and compensation for environmental damage and violations of environmental laws.
6. **Review and Appeals:** The Act outlines the process for review and appeals against NGT decisions. Appeals against NGT orders are heard by the Supreme Court of India.
7. **Fast-Track Process:** The NGT is designed to provide a fast-track process for resolving environmental disputes
8. **Role in Environmental Protection:**

Solid Waste Management Rules, 2016

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1. Objectives:

- Promote environmentally sound management of solid waste.
- Reduce the adverse environmental and public health impacts of improper waste disposal.
- Encourage segregation of waste at source and recycling.

2. Applicability:

Urban local body, outgrowths of urban areas, and all stakeholders involved in SWM.

3. Public Awareness and Education:

- The rules stress the importance of public awareness and education regarding the proper management of solid waste.
- Municipal authorities are encouraged to run awareness programs.

4. Impact:

- The Solid Waste Management Rules, 2016 have led to increased attention on proper **waste segregation and recycling.**
- The concept of **Extended Producer Responsibility** has been introduced to shift responsibility to manufacturers.
- Challenges include enforcement and creating infrastructure for waste management.

Key Provisions:

a) **Solid Waste Management Hierarchy:** - prioritizing waste minimization, segregation, recycling, and environmentally safe disposal.
b) **Waste Segregation at Source:** - biodegradable, non-biodegradable, and domestic hazardous waste.
c) **Responsibility of Generators:** - Producers, institutions, and residents are responsible for segregating waste and ensuring it is handed over to authorized waste collectors.
d) **Collection and Transportation:** - Local authorities and agencies are responsible for collecting, transporting, and ensuring safe disposal of waste. - Separate collection and transportation mechanisms are required for biodegradable and non-biodegradable waste.
e) **Waste Processing and Disposal:** - Biodegradable waste is to be processed through composting, vermicomposting, or other suitable methods. - Non-biodegradable waste is to be recycled, and residual waste is to be disposed of in a sanitary landfill.
f) **Extended Producer Responsibility (EPR):** - The rules introduce EPR, placing responsibility on producers to collect a certain percentage of the waste generated by their products.

g) **Hazardous Waste:** - Rules for handling and disposal of hazardous waste are incorporated within the framework. - Producers of such waste must manage it in accordance with hazardous waste rules.
h) **Sanitary Landfills:** - The rules set standards for the construction, operation, and closure of sanitary landfills. - They emphasize the need for leachate collection and treatment.
i) **Penalties and Enforcement:** - Penalties and fines are prescribed for non-compliance with the rules. - **Regulatory authorities, including the State Pollution Control Boards, are responsible for enforcement**.

Plastic Waste Management Rules, 2016

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1. Objectives:

- Regulate the manufacture, use, recycling, and disposal of plastic waste.
- Reduce the environmental impact of plastic waste by promoting responsible practices.
- Promote the **concept of "Extended Producer Responsibility" (EPR) in managing plastic waste.**

2. Applicability:

All persons involved in the generation, manufacture, storage, distribution, sale, use, collection, recycling, and processing of plastic and plastic waste.

3. Public Awareness and Education:

Emphasize the importance of public awareness and education regarding the hazards of plastic waste and the need for responsible disposal.

4. Impact:

The Plastic Waste Management Rules, 2016 have led to increased awareness about plastic waste management.

Efforts have been made to **reduce single-use plastics and promote recycling**.

Challenges include enforcement and the need for widespread behavior change.

5. Key Provisions:

a) **Plastic Waste Management Hierarchy:** - Prioritizing reduction, reuse, and recycling over disposal.

b) **Phase-Out of Non-Recyclable Multilayered Plastic:** - Manufacturers, sellers, and users of non-recyclable multilayered plastic are directed to phase it out within two years from the date of the notification of the rules.

c) **Responsibility of Generators:** - The rules place responsibility on individuals, businesses, and industries to segregate plastic waste at source.

d) **Extended Producer Responsibility (EPR):** - Manufacturers, importers, and brand owners are responsible for collecting and managing post-consumer plastic waste. They are required to establish a system for collecting plastic waste from the consumer and recycling it.

e) **Collection and Channelization:** - Local bodies and agencies are responsible for setting up a system for the collection, storage, transportation, and channelization of plastic waste.

f) **Waste Minimization:**

g) **Recycling:** - Plastic waste processing and recycling units are required to obtain authorization from the State Pollution Control Boards.

h) **Banning Plastic in Specific Areas:**.

i) **Penalties and Enforcement:** - Penalties and fines are prescribed for non-compliance with the rules. - Regulatory authorities, including the State Pollution Control Boards

e-Waste (Management) Rules, 2016

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1. Objectives:

- Ensure environmentally sound management of e-waste.
- Reduce the adverse environmental and health effects of improper e-waste disposal.
- Promote the concept of "Extended Producer Responsibility" (EPR) in managing e-waste.

2. Applicability:

- Every producer, consumer, bulk consumer, collection centers, dismantler, and recycler involved in the manufacture, sale, purchase, and processing of electrical and electronic equipment.

3. Public Awareness and Education:

- The rules emphasize the importance of public awareness and education about e-waste management.
- Producers and authorized agencies are encouraged to run awareness programs.

4. Impact:

- Awareness about the proper disposal and recycling of electronic waste.
- Extended Producer Responsibility
- Efficient collection systems and safe recycling facilities.

3. Key Provisions:

a) **Extended Producer Responsibility (EPR):** Producers must set up collection centers and take back mechanisms for e-waste generated by their products.
b) **Responsibility of Bulk Consumers:** - Bulk consumers of electrical and electronic equipment, such as government agencies and businesses, must ensure that their e-waste is handed over to authorized recyclers.
c) **Waste Minimization:** - Producers are encouraged to take measures to minimize e-waste generation at the design and manufacturing stages. - They must also promote environmentally friendly design and manufacturing processes.
d) **Collection and Transportation:** - Collection centers and authorized dismantlers are responsible for collecting and transporting e-waste to registered recycling facilities.
e) **Recycling and Dismantling:** - Recycling facilities and dismantlers must follow environmentally safe procedures for the recycling and disposal of e-waste.
f) **Registration and Authorization:** - Recycling facilities, dismantlers, and collection centers must register with the appropriate authorities and obtain authorization. - These facilities are subject to periodic inspections to ensure compliance.
g) **Penalties and Enforcement**

Hazardous Waste Management Rules, 2016

SHIV NADAR UNIVERSITY CHENNAI

1. Objectives:

- Ensure environmentally sound management of hazardous waste.
- Minimize the adverse impacts of hazardous waste on human health and the environment.
- Promote the concept of "Extended Producer Responsibility" (EPR) in managing hazardous waste.

2. Applicability:

- The rules apply to all persons involved in the generation, handling, treatment, and disposal of hazardous waste.

3. Impact:

- The Hazardous Waste Management Rules, 2016 have contributed to the safe and environmentally sound management of hazardous waste in India.
- They have promoted the concept of Extended Producer Responsibility, encouraging producers to take responsibility for waste generated by their products.
- **Challenges include effective monitoring and enforcement, especially in small-scale industries**.

3. Key Provisions:

- **Hazardous Waste Categories:** - The rules classify hazardous waste into categories based on its properties, including toxicity, persistence, and bioaccumulation potential. - Different categories require specific handling and disposal methods.
- **Extended Producer Responsibility (EPR):** - Producers of products must establish a system for the collection and proper disposal of waste generated by their products.
- **Treatment, Storage, and Disposal Facilities (TSDFs):** - The rules set standards for the establishment and operation of TSDFs, where hazardous waste is treated, stored, and disposed of. - TSDFs must obtain authorization from regulatory authorities.
- **Manifest System:** - A manifest system is introduced for the tracking and movement of hazardous waste. - Transporters, generators, and operators of TSDFs must use manifest forms to record waste movement.
- **Hazardous Waste Handling and Transportation:** - The rules establish guidelines for the safe handling, packaging, and transportation of hazardous waste. –
- **Waste Minimization and Segregation:** - Hazardous waste generators are encouraged to minimize waste generation and segregate hazardous waste at the source. - They are also required to maintain records of waste generated.
- **Public Awareness and Reporting:** - The rules emphasize public awareness regarding hazardous waste and the reporting of accidents or spillage.
- **Penalties and Enforcement:** - Penalties and fines are prescribed for non-compliance with the rules. - Regulatory authorities, including the State Pollution Control Boards, are responsible for enforcement.

Coastal Regulation Zone (CRZ) Notification, 2019

Objectives:

- Conserve and protect the coastal environment, its unique ecosystems, and wildlife.
- Regulate activities in coastal areas to minimize environmental degradation.
- Balance development with environmental conservation and sustainable livelihoods for coastal communities.

Key Provisions:

1. **Classification of Coastal Areas:** No Development Zone (NDZ), the CRZ-I, CRZ-II, and CRZ-III areas.
2. **No Development Zone (NDZ):** The NDZ is the area close to the shoreline.
3. **Regulated Activities:** The CRZ 2019 outlines activities permitted in the different coastal zones, including construction, tourism, and infrastructure development.
4. **CRZ-I Areas:** These areas have the highest degree of protection and are mainly ecologically sensitive areas.
5. **CRZ-II Areas:** CRZ-II areas include developed areas with relatively high population density.
6. **CRZ-III Areas:** These are rural and semi-urban areas with relatively lower population density.
7. **Promotion of Sustainable Development:** green infrastructure, conservation of wetlands, and the protection of mangrove forests.
8. **Penalties and Enforcement:** penalties for violations, which may include fines and imprisonment. Enforcement is carried out by the State Coastal Zone Management Authorities.